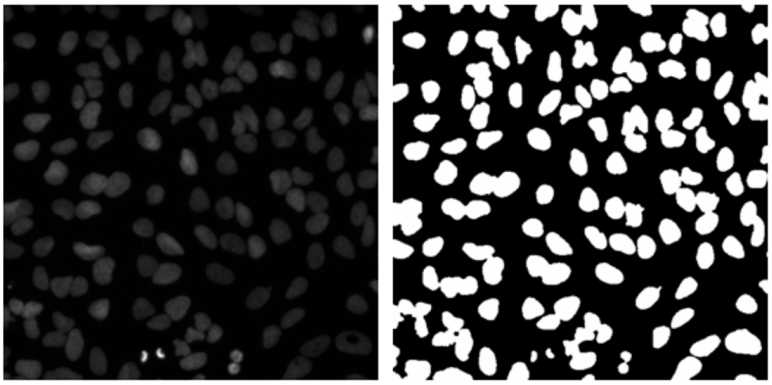
Tutorial Questions | Week 7

COSC2779 - Deep Learning

This tutorial is aimed at reviewing practical methodology in developing CNN. Please try the questions before you join the session. The question below has similar structure to the case study questions that you can expect in the end semester test.

1. You have been hired by a provider of pathology services in Victoria to solve one of their challenges in dealing with cell images: determining which parts of a microscope image corresponds to which individual cells. In deep learning, image segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics. In your case, you want to locate the cells and their boundaries in microscopic images. In other words, this is a classification task where each pixel of the target image is labeled as 0 (this pixel is not part of a cell) or 1 (this pixel is part of a cell).

   Here is an examples of an input image and the corresponding target image:

   The pathology company has collected 10,000 images from microscopes and gave them to you with annotations made by their doctors. The doctors have labeled each image as shown in the above figure, where all the pixels that belong to cells are labeled 1 and the other pixels that belong to the background are labelled 0. Images have been taken from three types of microscopes: A (5,000 images), B (2,500 images) and C (2,500 images). The doctors who hired you would like to use your algorithm on images from microscope C.

   (a) Explain how you would split this dataset into train, val and test sets. Give the exact percentage split, and give reasons to your choices.

   > **Solution:** Split has to be roughly 90,5,5. val and test set have to be the same (contain images from C). There should be C images in the training as well, more than in the test/valset.

   (b) Can you augment this dataset? If yes, give only 3 distinct methods you would use. If no, explain why (give only 2 reasons).

   > **Solution:** Those methods would work for augmentation based on the images shown above. -cropping -adding random noise -changing contrast, blurring. -flip -rotate

   (c) The first step in building the system is to identify the performance measures, What performance measure would you select and why?

   > **Solution:** pixel wise measure
   >
   > A performance measure that can handle class imbalance: e.g. F1 score
   >
   > Number of white pixels in the label image is different to the number of black pixels.

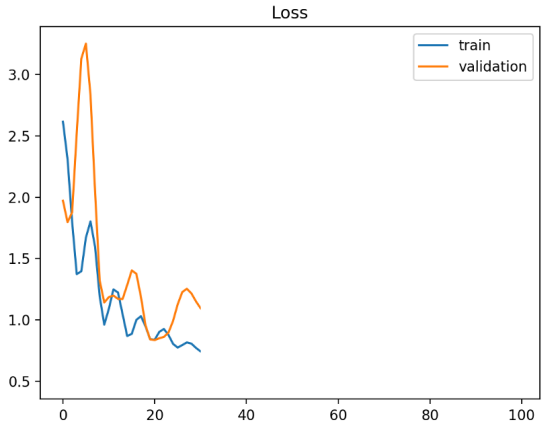
(d) You have decided to use a CNN based approach to solve this problem, What type of base network architecture would you use? Why?

> **Solution:** Encoder decoder type network like U-net. The most popular architecture for image segmentation is the encode decoder architecture which allows local to global information aggregation. U-net is the most popular architecture which is known to work well for medical image segmentation where there is relatively small amount of data.

(e) What will be a suitable cost function to train the model?

> **Solution:** A loss that is suitable for binary classification and can account for class imbalance.
>
> e.g: Weighted binary cross entropy.
>
> $$\mathcal{L} = -\sum_{i=1}^{N}\sum_{j=1}^{H\times W} \alpha_1 y_i \log \hat{y}_i + \alpha_0 \left(1 - y_i\right) \log\left(1 - \hat{y}_i\right)$$

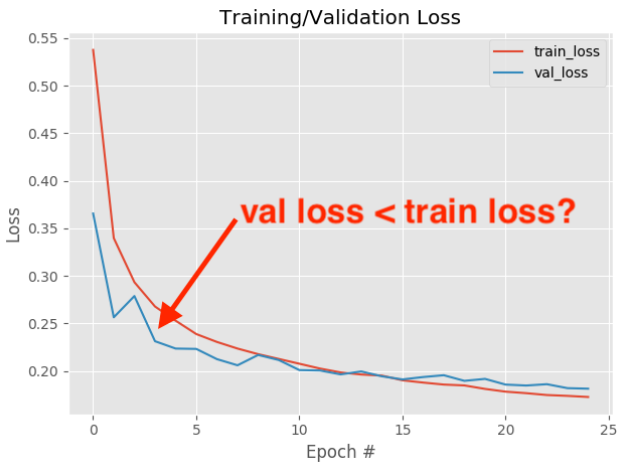
(f) While training the model you observed the following loss curve:



What are the possible alteration you might try?

> **Solution:** Both validation and training losses are oscillating. Therefore you might consider changing the learning rate.

(g) You have now fixed the issues with your model and the best learning curve you obtained is as follows:



How would you explain, Why the validation loss lower than training loss? what would you do?

Solution: Reason 1: Regularization applied during training, but not during validation/testing

Reason 2: Training loss is measured during each epoch while validation loss is measured after each epoch

Reason 3: The validation set may be easier than the training set (or there may be leaks)